

## Gabriel Stoltz

French nationality.  
Born 27th may 1979.  
Married, 2 children

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### Positions & Education

|                  |                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
| September 2022-  | Professor at Ecole polytechnique (part time)                                                                             |
| June 2016-       | Professor at Ecole des Ponts ParisTech                                                                                   |
| June 2012        | Habilitation thesis defended                                                                                             |
| September 2008 - | Researcher at CERMICS (Ecole des Ponts ParisTech, France) & MATHERIALS project, Inria                                    |
| 2007-2008        | Post-doc at IMPMC, University Paris 6<br>Numerical methods for quantum thermal transport (supervised by Francesco Mauri) |
| 2004-2007        | PhD in applied mathematics<br>"Some mathematical methods in molecular simulation"<br>Advisor: Eric Cancès, CERMICS       |
| 2002-2004        | Ecole Nationale des Ponts et Chaussées, Marne-La-Vallée<br>MS in Numerical Analysis, Université Paris VI                 |
| 1999-2002        | Ecole polytechnique, Paris<br>specialization in Mathematics                                                              |

### Scientific Experience

|                           |                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2008<br>(1.5 month)       | Junior program "Computational Mathematics"<br>(HIM, Bonn, Germany)                                                                                                                                   |
| 2005<br>(3 months)        | Program "Bridging time and length scales in materials science and biophysics"<br>(IPAM, UCLA, Los Angeles, USA)                                                                                      |
| 2004<br>(6 weeks)         | Summer School CEMRACS (Marseille, France).<br><i>Field of research:</i> Conformational molecular dynamics in drug design. <i>Supervisor:</i> Eric Cancès.                                            |
| 2003 - 2004<br>(8 months) | Training period in industry (CEA, French Atomic Energy Authority).<br><i>Field of research:</i> Upscaling issues in molecular dynamic simulations of shock waves. <i>Supervisor:</i> Claude Le Bris. |
| 2002<br>(3 months)        | Final training period of the Ecole polytechnique, at ETH Zürich.<br><i>Field of research:</i> Hardy Spaces and some applications to the Euler equation. <i>Supervisor:</i> Tristan Rivière.          |

## Prizes and awards

|      |                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 2020 | Chevalier dans l'ordre des Palmes académiques                                                                         |
| 2008 | PhD prize from Ecole des Ponts                                                                                        |
| 2002 | Rivot medal from the French Academy of Sciences (which honors academic performance in studies at Ecole polytechnique) |

## Languages

|         |        |
|---------|--------|
| English | fluent |
| German  | fluent |

## Computer Skills

|                      |               |
|----------------------|---------------|
| Programing languages | C/C++, Python |
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## Books and book chapters

- T. LELIÈVRE, M. ROUSSET AND G. STOLTZ, Free Energy Computations: A Mathematical Perspective, Imperial College Press (2010)
- G. GIACOMIN, S. OLLA, E. SAADA, H. SPOHN, G. STOLTZ (editors), Stochastic Dynamics Out Of Equilibrium, Springer Proceedings in Mathematics & Statistics, volume 282 (2019)
- C. CHIPOT, P. GKEKA, T. LELIÈVRE AND G. STOLTZ, Equilibrium and nonequilibrium methods for free-energy calculations with molecular dynamics, *Reference Module in Chemistry, Molecular Sciences and Chemical Engineering*, **3**, pages 384-400, Elsevier (2024)

## Publications

- (1) G. BRIGATI, G. STOLTZ, A. Q. WANG AND L. WANG, Explicit convergence rates of underdamped Langevin dynamics under weighted and weak Poincaré–Lions inequalities, accepted for publication in *Annals of Applied Probability* (2026)
- (2) A. KARONI, R. RAJPAL, B. LEIMKUHLE AND G. STOLTZ, Adaptive momentum and nonlinear damping for neural network training, presented at *ICML* (2026)
- (3) R. GASTALDELLO, G. STOLTZ AND U. VAES, Dynamical reweighting for estimation of fluctuation formulas, accepted for publication in *Multiscale Model. Sim.* (2026)
- (4) F.-Z. AKHYAR, W. ZHANG, G. STOLTZ, C. SCHÜTTE, Generative modeling of conditional probability distributions on the level-sets of collective variables, accepted for publication in *Communications in Applied Mathematics and Computational Science* (2026)
- (5) N. BLASSEL, T. LELIÈVRE AND G. STOLTZ, Shape optimization of metastable states, accepted for publication in *Mathematical Models and Methods in Applied Sciences* (2026)
- (6) C. SCHÖNLE, D. CARBONE, M. GABRIÉ, T. LELIÈVRE AND G. STOLTZ, Efficient Monte-Carlo sampling of metastable systems using non-local collective variable updates, *J. Chem. Phys.* **164** 154107 (2026)
- (7) S. DARSHAN, A. IACOBUCCI, S. OLLA AND G. STOLTZ,  $\beta$ -FPUT chains under time-periodic forcing, *J. Stat. Phys.* **193**, 37 (2026)
- (8) G. PAVLIOTIS, R. SPACEK, G. STOLTZ AND U. VAES, Neural network approaches for variance reduction in fluctuation formulas, *SIAM/ASA Journal of Uncertainty Quantification* **14**(1) 221-255 (2026)
- (9) M. CHAK, T. LELIÈVRE, G. STOLTZ AND U. VAES, Optimal importance sampling for overdamped Langevin dynamics, *Bernoulli* **32**(3) 1827-1852 (2026)
- (10) T. LELIÈVRE, R. SANTET AND G. STOLTZ, Improving sampling by modifying the effective diffusion, *J. Comput. Phys.* **541** 114313 (2025)
- (11) A. CASTELLANO, R. BÉJAUD, P. RICHARD, O. NADEAU, C. DUVAL, G. GENESTE, G. ANTONIUS, J. BOUCHET, A. LEVITT, G. STOLTZ AND F. BOTTIN, Machine learning assisted canonical sampling (MLACS), *Computer Physics Communications* **316** 109730 (2025)
- (12) T. LELIÈVRE, G. A. PAVLIOTIS, G. ROBIN, R. SANTET AND G. STOLTZ, Optimizing the diffusion of overdamped Langevin dynamics, accepted for publication in *Mathematics of Computation* (2025)
- (13) G. BRIGATI AND G. STOLTZ, How to construct decay rates for kinetic Fokker–Planck equations?, *SIAM J. Math. Anal.* **57**(4), 3587-3622 (2025)
- (14) P. MONMARCHÉ, R. SPACEK AND G. STOLTZ, Transient subtraction: A control variate method for computing transport coefficients, *J. Stat. Phys.* **192**, 53 (2025)
- (15) C. SCHÖNLE, M. GABRIÉ, T. LELIÈVRE AND G. STOLTZ, Sampling metastable systems using collective variables and Jarzynski–Crooks paths, *J. Comput. Phys.* **527**, 113806 (2025)

- (16) T. LELIÈVRE, R. SANTET AND G. STOLTZ, Unbiasing Hamiltonian Monte Carlo algorithms for a general Hamiltonian function, *Found. Comput. Math.* **26**, 1-74 (2026)
- (17) T. LELIÈVRE, T. PIGEON, G. STOLTZ AND W. ZHANG, Analyzing multimodal probability measures with autoencoders, *J. Phys. Chem. B* **128**(11) 2607-2631 (2024)
- (18) N. BLASSEL AND G. STOLTZ, Fixing the flux: A dual approach to computing transport coefficients, *J. Stat. Phys.* **191**, 17 (2024)
- (19) I. SEKKAT AND G. STOLTZ, Removing the mini-batching error in Bayesian inference using Adaptive Langevin dynamics, *J. Mach. Learn. Res.* **24**(329), 1-58 (2023)
- (20) A. DURMUS, A. ENFROY, E. MOULINES AND G. STOLTZ, Uniform minorization condition and convergence bounds for discretizations of kinetic Langevin dynamics, *Ann. Inst. H. Poincaré Probab. Statist.* **61**(1), 629-664 (2025)
- (21) K. KARONI, B. LEIMKUHLE AND G. STOLTZ, Friction-adaptive descent: A family of dynamics-based optimization methods, *J. Comput. Dyn.* **10**(4), 450-484 (2023)
- (22) R. SPACEK AND G. STOLTZ, Extending the regime of linear response with synthetic forcings, *Multiscale Model. Sim.* **21**(4) 1602-1643 (2023)
- (23) Z. BELKACEMI, M. BIANCIOTTO, H. MINOUX, T. LELIÈVRE, G. STOLTZ AND P. GKEKA, Autoencoders for dimensionality reduction in molecular dynamics: collective variable dimension, biasing and transition states, *J. Chem. Phys.* **159**, 024122 (2023)
- (24) T. PIGEON, G. STOLTZ, M. CORRAL-VALERO, A. ANCIAUX-SEDRAKIAN, M. MOREAUD, T. LELIÈVRE AND P. RAYBAUD, Computing surface reaction rates by Adaptive Multilevel Splitting combined with machine learning and ab initio molecular dynamics, *J. Chem. Theory Comput.* **19**(12), 3538-3550 (2023)
- (25) T. LELIÈVRE, G. ROBIN, I. SEKKAT, G. STOLTZ, AND G. VICTORINO CARDOSO, Generative methods for sampling transition paths in molecular dynamics, *ESAIM Proc.* **73**, 238-256 (2023)
- (26) T. HOANG NGOC MINH, G. STOLTZ, AND B. ROTENBERG, Frequency and field-dependent response of confined electrolytes from Brownian dynamics simulations, *J. Chem. Phys.* **158**, 104103 (2023)
- (27) G. A. PAVLIOTIS, G. STOLTZ, U. VAES, Mobility estimation for Langevin dynamics using control variates, *Multiscale Model. Sim.* **21**(2), 680-715 (2023)
- (28) A. CASTELLANO, F. BOTTIN, J. BOUCHET, A. LEVITT AND G. STOLTZ, Ab initio canonical sampling based on variational inference, *Phys. Rev. B* **106**, L161110 (2022)
- (29) P. PLECHAC, G. STOLTZ AND T. WANG, Martingale product estimators for sensitivity analysis in computational statistical physics, *IMA J. Numer. Anal.* **43**(6), 3430-3477 (2023)
- (30) T. LELIÈVRE, G. STOLTZ AND W. ZHANG, Multiple projection MCMC algorithms on submanifolds, *IMA J. Numer. Anal.* **43**(2), 737-788 (2023)
- (31) E. CAMRUD, D. P. HERZOG, G. STOLTZ, M. GORDINA, Weighted  $L^2$ -contractivity of Langevin dynamics with singular potentials, *Nonlinearity* **35**(2), 998-1035 (2022)
- (32) E. BERNARD, M. FATHI, A. LEVITT AND G. STOLTZ, Hypocoercivity with Schur complements, *Annales Henri Lebesgue* **5**, 523-557 (2022)
- (33) Z. BELKACEMI, P. GKEKA, T. LELIÈVRE AND G. STOLTZ, Chasing collective variables using autoencoders and biased trajectories, *J. Chem. Theory Comput.* **18**(1), 59-78 (2022)
- (34) A. IACOBUCCI, S. OLLA, G. STOLTZ, Thermo-mechanical transport in rotor chains, *J. Stat. Phys.* **183**, 26 (2021)
- (35) G. A. PAVLIOTIS, G. STOLTZ, U. VAES, Scaling limits for the generalized Langevin equation, *Journal of Nonlinear Science* **31**, no. 8 (2021)
- (36) D. CHAFAÏ, G. FERRÉ AND G. STOLTZ, Coulomb gases under constraint: some theoretical and numerical results, *SIAM J. Math. Anal.* **53**(1), 181-220 (2021)

- (37) G. FERRÉ AND G. STOLTZ, Large deviations of the empirical measure of diffusions in weighted topologies with applications, *Electron. J. Probab.* **25**, 121 (2020)
- (38) P. PLECHAC, G. STOLTZ AND T. WANG, Convergence of the likelihood ratio method for linear response of non-equilibrium stationary states, *M2AN* **55**, S593-S623 (2021)
- (39) G. FERRÉ, M. ROUSSET AND G. STOLTZ, More on the long time stability of Feynman-Kac semi-groups, *Stoch. PDE* **9**(3), 630-673 (2021)
- (40) P. GKEKA, G. STOLTZ, A. BARATI FARIMANI, Z. BELKACEMI, M. CERIOTTI, J. CHODERA, A. DINNER, A. FERGUSON, J.-B. MAILLET, H. MINOUX, C. PETER, F. PIETRUCCHI, A. SILVEIRA, A. TKATCHENKO, Z. TRSTANOVA, R. WIEWIORA, T. LELIÈVRE, Machine learning force fields and coarse-grained variables in molecular dynamics: application to materials and biological systems, *J. Chem. Theory Comput.* **16**(8), 4757-4775 (2020)
- (41) L. PILLAUD-VIVIEN, F. BACH, T. LELIÈVRE, A. RUDI AND G. STOLTZ, Statistical estimation of the Poincaré constant and application to sampling multimodal distributions, *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, *PMLR* **108**, 2753-2763 (2020)
- (42) B. LEIMKUHLER, M. SACHS AND G. STOLTZ, Hypocoercivity properties of adaptive Langevin dynamics, *SIAM J. Appl. Math.* **80**(3), 1197-1222 (2020)
- (43) E. CANCÈS, L. CAO AND G. STOLTZ, A reduced Hartree-Fock model of slice-like defects in the Fermi sea, *Nonlinearity* **33**(1), 156-195 (2020)
- (44) T. LELIÈVRE, M. ROUSSET AND G. STOLTZ, Hybrid Monte Carlo methods for sampling probability measures on submanifolds, *Numer. Math* **143**(2), 379-421 (2019)
- (45) G. FERRÉ AND G. STOLTZ, Error estimates on ergodic properties of discretized Feynman-Kac semi-groups, *Numer. Math* **143**(2), 261-313 (2019)
- (46) J. ROUSSEL AND G. STOLTZ, A perturbative approach to control variates in molecular dynamics, *Multiscale Model. Sim.* **17**(1), 552-591 (2019)
- (47) G. STOLTZ AND E. VANDEN-ELJNDEN, Longtime convergence of the Temperature-Accelerated Molecular Dynamics Method, *Nonlinearity* **31**(8), 3748-3769 (2018)
- (48) G. FORT, B. JOURDAIN, T. LELIÈVRE AND G. STOLTZ, Convergence and efficiency of adaptive importance sampling techniques with partial biasing, *J. Stat. Phys.* **171**(2), 220-268 (2018)
- (49) G. STOLTZ AND Z. TRSTANOVA, Langevin dynamics with general kinetic energies, *Multiscale Model. Sim.* **16**(2), 777-806 (2018)
- (50) G. FAURE AND G. STOLTZ, Stable and accurate schemes for smoothed dissipative particle dynamics, *Appl. Math. Mech.-Engl.* **39**(1), 83-102 (2018)
- (51) A. IACOBUCCI, S. OLLA AND G. STOLTZ, Convergence rates for nonequilibrium Langevin dynamics, *Ann. Math. Quebec* **43**(1), 73-98 (2019)
- (52) J. ROUSSEL AND G. STOLTZ, Spectral methods for Langevin dynamics and associated error estimates, *M2AN* **52**(3), 1051-1083 (2018)
- (53) P. TERRIER, M. ATHÈNES, T. JOURDAN, G. ADJANOR AND G. STOLTZ, Cluster dynamics modelling of materials: a new hybrid deterministic/stochastic coupling approach, *J. Comput. Phys.* **350**, 280-295 (2017)
- (54) G. STOLTZ, Stable schemes for dissipative particle dynamics with conserved energy, *J. Comput. Phys.* **340**, 451-469 (2017)
- (55) E. CANCÈS, A. LEVITT, G. PANATI AND G. STOLTZ, Robust determination of maximally-localized Wannier functions, *Phys. Rev. B* **95**, 075114 (2017)
- (56) A. LESAGE, T. LELIÈVRE, G. STOLTZ AND J. HÉNIN, Smoothed biasing forces yield unbiased free energies with the extended-system adaptive biasing force method, *J. Phys. Chem. B* **121**(15), 3676-3685 (2017)

- (57) M. FATHI AND G. STOLTZ, Improving dynamical properties of stabilized discretizations of overdamped Langevin dynamics, *Numer. Math.* **136**(2), 545-602 (2017)
- (58) G. FAURE, J. ROUSSEL, J.-B. MAILLET AND G. STOLTZ, Size consistency in Smoothed Dissipative Particle Dynamics, *Phys. Rev. E* **94**, 043305 (2016)
- (59) T. JOURDAN, G. STOLTZ, F. LEGOLL AND L. MONASSE, An accurate scheme to solve cluster dynamics equations using a Fokker-Planck approach, *Comput. Phys. Commun.* **207**, 170-178 (2016)
- (60) E. CANCÈS, D. GONTIER AND G. STOLTZ, A mathematical analysis of the  $\text{GW}^0$  method for computing electronic excited state energies of molecules, *Rev. Math. Phys.* **28**(4), 1650008 (2016)
- (61) T. LELIÈVRE AND G. STOLTZ, Partial differential equations and stochastic methods in molecular dynamics, *Acta Numerica* **25**, 681-880 (2016)
- (62) S. REDON, G. STOLTZ AND Z. TRSTANOVA, Error Analysis of Modified Langevin Dynamics, *J. Stat. Phys.* **164**(4), 735-771 (2016)
- (63) I. G. TEJADA, L. BROCHARD, T. LELIÈVRE, G. STOLTZ, F. LEGOLL AND E. CANCÈS, Coupling a reactive potential with a harmonic approximation for atomistic simulations of material failure, *Computer Methods in Applied Mechanics and Engineering* **305**, 422-440 (2016)
- (64) A.-A. HOMMAN, J.-B. MAILLET, J. ROUSSEL AND G. STOLTZ, New parallelizable schemes for integrating the Dissipative Particle Dynamics with Energy Conservation, *J. Chem. Phys.* **144**, 024112 (2016)
- (65) G. FORT, B. JOURDAIN, T. LELIÈVRE AND G. STOLTZ, Self-Healing Umbrella Sampling: Convergence and efficiency, *Stat. Comput.* **27**(1), 147-168 (2017)
- (66) G. FERRÉ, J.-B. MAILLET AND G. STOLTZ, Permutation-invariant distance between atomic configurations, *J. Chem. Phys.* **143** 104114 (2015)
- (67) H. SPOHN AND G. STOLTZ, Nonlinear fluctuating hydrodynamics in one dimension: the case of two conserved fields, *J. Stat. Phys.* **160**(4), 861-884 (2015)
- (68) B. LEIMKUHLE, CH. MATTHEWS AND G. STOLTZ, The computation of averages from equilibrium and nonequilibrium Langevin molecular dynamics, *IMA J. Numer. Anal.* **36**(1), 13-79 (2016)
- (69) R. JOUBAUD, G. PAVLIOTIS AND G. STOLTZ, Langevin dynamics with space-time periodic nonequilibrium forcing, *J. Stat. Phys.* **158**(1), 1-36 (2015)
- (70) M. FATHI, A.-A. HOMMAN AND G. STOLTZ, Error analysis of the transport properties of Metropolized schemes, *ESAIM Proc.* **48**, 341-363 (2015)
- (71) G. FORT, B. JOURDAIN, E. KUHN, T. LELIÈVRE AND G. STOLTZ, Convergence of the Wang-Landau algorithm, *Math. Comput.* **84**(295), 2297-2327 (2015)
- (72) G. FAURE, J.-B. MAILLET AND G. STOLTZ, Local density dependent potential for compressible mesoparticles, *J. Chem. Phys.* **140** 114105 (2014)
- (73) G. FORT, B. JOURDAIN, E. KUHN, T. LELIÈVRE AND G. STOLTZ, Efficiency of the Wang-Landau algorithm: a simple test case, *Appl. Math. Res. Express* 2014(2), 275-311 (2014)
- (74) L. CAO, M.-C. MARINICA, G. STOLTZ, T. LELIÈVRE AND M. ATHÈNES, Using Bayes formula to average the biasing forces and recover free energies in adaptive Monte Carlo simulations, *J. Chem. Phys.* 104108 (2014)
- (75) A.-A. HOMMAN, E. BOURASSEAU, G. STOLTZ, P. MALFREYT, L. STRAFELLA AND A. GHOUFI, Surface tension of spherical drops from surface of tension, *J. Chem. Phys.*, **140** (2014) 034110
- (76) M. DOBSON, F. LEGOLL, T. LELIÈVRE AND G. STOLTZ, Derivation of Langevin dynamics in a nonzero background flow field, *M2AN* **47** (2013) 1583-1626 (2013)
- (77) E. CANCÈS AND G. STOLTZ, A mathematical formulation of the random phase approximation for crystals, *Ann. I. H. Poincaré-An.* **29**(6) (2012) 887-925



- (78) C. BERNARDIN AND G. STOLTZ, Anomalous diffusion for a class of systems with two conserved quantities, *Nonlinearity* **25** (2012) 1099-1133
- (79) R. JOUBAUD AND G. STOLTZ, Nonequilibrium shear viscosity computations with Langevin dynamics, *Multiscale Model. Sim.* **10** (2012) 191-216
- (80) T. LELIÈVRE, M. ROUSSET AND G. STOLTZ, Langevin dynamics with constraints and computation of free energy differences, *Math. Comput.* **81** (2011) 2071-2125
- (81) N. CHOPIN, T. LELIÈVRE AND G. STOLTZ, Free energy methods for efficient exploration of mixture posterior densities, *Stat. Comput.* **22**(4) (2012) 897-916
- (82) J.-B. MAILLET, G. VALLVERDU, N. DESBIENS AND G. STOLTZ, Molecular Simulations of Shock to Detonation Transition in Nitromethane, *Europhys. Lett.* **96** (2011) 68007
- (83) A. IACOBUCCI, F. LEGOLL, S. OLLA, G. STOLTZ, Negative thermal conductivity of chains of rotors with mechanical forcing, *Phys. Rev. E* **84** (2011) 061108
- (84) E. BOURASSEAU, J.-B. MAILLET, N. DESBIENS AND G. STOLTZ, Microscopic calculations of Hugoniot curves of neat TATB and of its detonation products, *J. Phys. Chem. A* **115** (39) (2011) 10729-10737
- (85) A. IACOBUCCI, F. LEGOLL, S. OLLA, G. STOLTZ, Thermal conductivity of the Toda lattice with conservative noise, *J. Stat. Phys.* **140**(2) (2010) 336-348
- (86) B. M. DICKSON, F. LEGOLL, T. LELIÈVRE, G. STOLTZ, P. FLEURAT-LESSARD, Free energy calculations: An efficient adaptive biasing potential method, *J. Phys. Chem. B* **114**(17) (2010) 5823-5830
- (87) C. BROUDER, G. PANATI AND G. STOLTZ, Gell-Mann and Low formula for degenerate unperturbed states, *Ann. I. H. Poincaré-Phy* **10**(7) (2010) 1285-1309
- (88) C. BROUDER, G. PANATI AND G. STOLTZ, Many-body Green function of degenerate systems, *Phys. Rev. Lett.* **103** (2009) 230401
- (89) G. STOLTZ, N. MINGO AND F. MAURI, Reducing the thermal conductivity of carbon nanotubes below the random isotope limit, *Phys. Rev. B* **80** (2009) 113408
- (90) J.-B. MAILLET, E. BOURASSEAU, L. SOULARD, J. CLEROUIN, G. STOLTZ, Constant entropy sampling and release waves of shock compressions, *Phys. Rev. E* **80** (2009) 021135
- (91) G. STOLTZ, M. LAZZERI AND F. MAURI, Thermal transport in isotopically disordered carbon nanotubes, *J. Phys.:Cond. Matter* **21** (2009) 245302
- (92) E. CANCÈS, G. STOLTZ, G. SCUSERIA, V. STAROVEROV AND E. DAVIDSON, Local exchange potentials for electronic structure calculations, *MathS In Action* **2** (2009) 1-42
- (93) J.B. MAILLET AND G. STOLTZ, Sampling constraints in average: The example of Hugoniot curves, *Appl. Math. Res. Express* **2008** abn004 (2009)
- (94) C. BROUDER, G. STOLTZ AND G. PANATI, Adiabatic approximation, Gell-Mann and Low theorem and degeneracies: A pedagogical example, *Phys. Rev. A* **72** (2008) 042102
- (95) T. LELIÈVRE, M. ROUSSET AND G. STOLTZ, Long-time convergence of an adaptive biasing force method, *Nonlinearity* **21** (2008) 1155-1181
- (96) J.-B. MAILLET, L. SOULARD AND G. STOLTZ, A reduced model for shock and detonation waves. II. The reactive case, *Europhys. Lett.* **78**(6) (2007) 68001
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- (99) E. CANCÈS, F. LEGOLL AND G. STOLTZ, Theoretical and numerical comparison of some sampling methods, *M2AN* **41**(2) (2007) 351-390

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- (101) T. LELIÈVRE, M. ROUSSET AND G. STOLTZ, Computation of free energy differences through nonequilibrium stochastic dynamics: the reaction coordinate case, *J. Comput. Phys.* **222**(2) (2007) 624-643.
- (102) G. STOLTZ, A reduced model for shock and detonation waves. I. The inert case, *Europhys. Lett.* **76**(5) (2006) 849-855.
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- (104) E. CANCÈS, M. LEWIN AND G. STOLTZ, The electronic ground state energy problem: a new reduced density matrix approach, *J. Chem. Phys.* **125** (2006) 064101.
- (105) M. ROUSSET AND G. STOLTZ, An interacting particle system approach for molecular dynamics, *J. Stat. Phys.* **123**(6) (2006) 1251-1272.
- (106) G. STOLTZ, Shock waves in an augmented one-dimensional chain, *Nonlinearity* **18** (2005) 1967-1985.



## Submitted papers and technical reports

- L. CARILLO, T. LELIÈVRE, T. NORMAND, G. STOLTZ, U. VAES, The narrow escape problem in arbitrary dimension, *HAL preprint* **05725647** (2026)
- T. PIGEON, G. STOLTZ AND T. LELIÈVRE, Approximating committor functions: Objective functions and training data sampling, *HAL preprint* **05704510** (2026)
- R. GASTALDELLO, G. A. PAVLIOTIS, G. STOLTZ AND U. VAES, On the diffusive-mean field limit of a kinetic weakly interacting particle system, *HAL preprint* **05697822** (2026)
- N. BLASSEL, L. CARILLO, S. DARSHAN, R. GASTALDELLO, A. IACOBUCCI, E. MARINI, R. SANTET, X. SHANG, G. STOLTZ AND U. VAES, Mathematical analysis and numerical methods for the computation of transport coefficients in molecular dynamics, *HAL preprint* **05619533** (2006)
- N. BLASSEL, T. LELIÈVRE AND G. STOLTZ, Sharp low-temperature spectral asymptotics for reversible diffusions in temperature-dependent domains, *HAL preprint* **04917900** (2025)
- S. DARSHAN, A. EBERLE AND G. STOLTZ, Sticky coupling as a control variate for sensitivity analysis, *HAL preprint* **04708794** (2024)
- E. CAMRUD, A. O. DURMUS, P. MONMARCHÉ AND G. STOLTZ, Second order quantitative bounds for unadjusted generalized Hamiltonian Monte Carlo, *HAL preprint* **04574037** (2024)
- T. LELIÈVRE, M. RACHID AND G. STOLTZ, A spectral approach to the narrow escape problem in the disk, *HAL preprint* **04539840** (2024)
- G. STOLTZ, Computational statistical physics and hypocoercivity, *HAL preprint* **03482498** (2021)
- P. TERRIER AND G. STOLTZ, A mathematical justification of the finite time approximation of Becker–Döring equations by a Fokker–Planck dynamics, *HAL preprint* **02065570** (2019)

## Conference proceedings and related material

- G. STOLTZ, Error estimates and variance reduction for nonequilibrium stochastic dynamics. In A. Hinrichs, P. Kritzer and F. Pillichshammer (eds), *Monte Carlo and Quasi-Monte Carlo Methods (MCQMC 2022)*, Springer Proceedings in Mathematics & Statistics, vol **460**, pages 163-187 (2024)
- I. G. TEJADA, L. BROCHARD, G. STOLTZ, F. LEGOLL, T. LELIÈVRE AND E. CANCÈS, Combining a reactive potential with a harmonic approximation for molecular dynamics simulation of fracture, *IOP Journal of Physics: Conference Series* **574** (2015) 012041
- B. LEIMKUHLER AND G. STOLTZ, Sampling techniques for computational statistical physics, in *Encyclopedia of Applied and Computational Mathematics*, B. Engquist (Ed.) (Springer, 2012)
- G. STOLTZ, Calculation of ensemble averages, in *Encyclopedia of Applied and Computational Mathematics*, B. Engquist (Ed.) (Springer, 2012)
- G. STOLTZ, Computation of free energy differences, in *Encyclopedia of Applied and Computational Mathematics*, B. Engquist (Ed.) (Springer, 2012)
- E. CANCÈS, M. LEWIN AND G. STOLTZ, The microscopic origin of the macroscopic dielectric permittivity of crystals: A mathematical viewpoint, in *Numerical Analysis of Multiscale Computations*, B. Engquist, O. Runborg, Y.-H. R. Tsai. (Eds.), Lecture Notes in Computational Science and Engineering, Vol. 82 (2011) 87 - 125
- F. LEGOLL, T. LELIÈVRE AND G. STOLTZ, Some remarks on sampling methods in Molecular Dynamics, Proceedings of CANUM 2006, *ESAIM Proc* **22** (2008) 217-233

## Invited lectures and courses

- (Un)supervised learning with applications to molecular dynamics, 9h lecture, 7th edition of the Mini-school on mathematics for theoretical chemistry and physics (May 2025, Sorbonne Université, Paris, France)
- High Dimensional Sampling and Applications, 2h lecture + 1h seminar, MAC-MIGS tutorial (Edinburgh, November 2022)
- Constructing collective variables using Machine Learning and free energy biased simulations, 3h lecture + 3h hands-on session at AlgoSB Winter school 2021 (with Tony Lelièvre and Thomas Pigeon; November 2021)
- Sampling high-dimensional probability distributions and Bayesian learning, 6h lecture at doctoral school UM6P (Morocco, November 2019)
- From a microscopic description of matter to a macroscopic one on a computer: computational statistical physics, 6h lecture at the CIMPA Summer School on Multiscale Computational Methods and Error Control (IIT Kanpur, India, July 2017)
- A mathematical introduction to steady-state nonequilibrium systems, 2h lecture at the Spring school of the DFG collaborative research centre 1114 on Molecular Dynamics (Bad Belzig, Germany, April 2016)
- An introduction to molecular dynamics, 2h lecture at MOMAS meeting (CIRM, Marseille, November 2014)
- Molecular simulation: A mathematical introduction, 4h lecture at the School “Multi-scale and Multi-field Representations of Condensed Matter Behavior” (Pisa, November 2013)
- Molecular dynamics: a mathematical introduction, 4h lecture at the School “Longtime limits of stochastic models” (CIRM, Marseille, February 2013)

## Oral presentations (conferences and workshops)

- Nonequilibrium stochastic dynamics: Error estimates and variance reduction, plenary talk at Scicade 2026 (Edinburgh, United-Kingdom, July 2026)
- Autoencoders for free energy computations in molecular dynamics, AiChemist Paris School (Sanofi, Vitry-sur-Seine, France, March 2026)
- Error estimates and variance reduction for transport coefficients, workshop “CoMPASs: computational materials science and mathematics at the particle and atomistic scales” (ICMS, Edinburgh, UK, November 2025)
- Error estimates and variance reduction for transport coefficients, SCALES Conference 2025 (Mainz, Germany, September 2025)
- Error estimates in molecular dynamics, CECAM summer school “Sampling High-Dimensional Probability Measures with Applications in (Non)Equilibrium Molecular Dynamics and Statistics” (Birmingham, UK, July 2025)
- Hypocoercivity without changing the scalar product, Bernoulli program “New trends and applications around generalized Fokker-Planck operators” (Lausanne, Switzerland, July 2025)
- A mathematical analysis of autoencoders for (un)supervised dimensionality reduction, CECAM workshop “Data-driven, low-dimensional, and generative models for molecular and materials discovery and design” (UChicago Center in Paris, Paris, June 2025)
- Optimizing the diffusion for sampling with overdamped Langevin dynamics, workshop “Monte Carlo sampling: beyond the diffusive regime” (Newton Institute, Cambridge, UK, November 2024)

- Computational statistical physics and hypocoercivity, Workshop “The many facets of kinetic theory” (ICMS, Edinburgh, UK, September 2024)
- Error estimates in molecular dynamics, Workshop “Uncertainty Quantification in Molecular Simulation” (Magdeburg, Germany, August 2024)
- A mathematical analysis of autoencoders in the context of importance sampling, Workshop on “Synergies Between Mathematics, Data Science, and Molecular Simulations in Materials Science” (Birmingham, United-Kingdom, July 2024)
- A mathematical analysis of autoencoders for free energy calculations, Oberwolfach meeting “Constrained Dynamics, Stochastic Numerical Methods and the Modeling of Complex Systems” (Germany, May 2024)
- Enhanced sampling with autoencoders, CECAM workshop “Quantum2 on machine learning enhanced sampling” (Lausanne, Switzerland, November 2023)
- Finding collective variables with autoencoders, MOANSI annual meeting (Stuttgart, Germany, November 2023)
- Optimizing the diffusion for sampling with overdamped Langevin dynamics, program “Probabilistic sampling for physics” at Institut Pascal (Orsay, France, September 2023)
- Error estimates and variance reduction for nonequilibrium stochastic dynamics, workshop “Nonequilibrium molecular dynamics” (Birmingham, United-Kingdom, May 2023)
- Coarse-graining and efficiently sampling with autoencoders, Workshop “Rare event sampling” (BRIN center, University of Maryland, February 2023)
- Coarse-graining and efficiently sampling with autoencoders, MASIM ML & sampling workshop (Paris, December 2022)
- Reducing the mini-batching error in Bayesian inference using Adaptive Langevin dynamics, workshop “Machine-learning assisted scientific computing” (Paris, October 2022)
- Error estimates and variance reduction for nonequilibrium stochastic dynamics, plenary talk at MCQMC 2022 (Linz, Austria, July 2022)
- Numerical methods for the linear response of nonequilibrium stochastic dynamics, CNRS-ICL workshop (London, UK, July 2022)
- Coarse-graining molecular systems with autoencoders and adaptive sampling, NOMATEN International Conference on Materials Informatics (Warsaw, Poland, June 2022)
- Quantifying errors in the computation of transport coefficients, CECAM workshop “Numerical Techniques for Nonequilibrium Steady States” (Mainz, Germany, April 2022)
- Hypocoercivity without changing the scalar product, Workshop ANR EFI (Paris, France, October 2021)
- Removing the mini-batching error in Bayesian inference using Adaptive Langevin dynamics, Workshop “On Future Synergies for Stochastic and Learning Algorithms” (CIRM, France, October 2021)
- Computational statistical physics and hypocoercivity, Summer school “From kinetic equations to statistical mechanics” (St Jean de Monts, France, June 2021)
- Machine learning for coarse-graining molecular systems, MAC-MIGS afternoon on Mathematical Research Topics in Machine Learning (online, June 2021)
- Finding reaction coordinates with machine learning techniques for free energy computations, SIAM Materials (online, May 2021)
- Finding reaction coordinates with machine learning techniques for free energy computations, ERC Synergy workshop (online, February 2021)
- Hypocoercivity of Langevin-like dynamics with Schur complements, Bernoulli-IMS One World Symposium (online, August 2020)

- Hypocoercivity with Schur complements, Hypocoercivity workshop, Heilbronn Institute (Bristol, UK, March 2020)
- Removing the mini-batching error in large scale Bayesian sampling, Bayes 2020 (Gainesville, Florida, USA, January 2020)
- Error estimates in molecular dynamics, QuAMProcs meeting (Bordeaux, France, November 2019)
- Hybrid Monte Carlo methods for sampling on submanifolds, SciCADE 2019 (Innsbruck, Austria, July 2019)
- Convergence of Adaptive Langevin dynamics, ICIAM 2019 (Valencia, Spain, July 2019)
- Molecular dynamics by applied mathematicians, Rencontres prospectives RFCT, (Nantes, France, June 2019)
- Convergence of Adaptive Langevin (and other dynamics) using hypocoercivity, CIB workshop “Computational mathematics for model reduction and predictive modelling in molecular and complex systems” (Lausanne, Switzerland, May 2019)
- Some mathematical challenges in molecular dynamics, CECAM workshop “Microscopic simulations: forecasting the next two decades” (Toulouse, France, April 2019)
- Reducing error in molecular dynamics, Workshop Cambridge/Labex MMCD (Ecole des Ponts, France, April 2019)
- Molecular simulation and the numerical microscope, Journées scientifiques Inria 2018 (Bordeaux, France, June 2018)
- Efficient sampling of nonequilibrium systems, Focus Program on Nanoscale Systems and Coupled Phenomena: Mathematical Analysis, Modeling, and Applications (Fields institute, Toronto, Canada, May 2018)
- Parametrizing coarse-grained molecular systems from ab-initio computations: some elements, IPAM workshop “Collective Variables in Classical Mechanics” (Los Angeles, USA, October 2016)
- Error estimates for transport coefficients in molecular dynamics, MMM16 (Dijon, France, October 2016)
- A mathematical study of the  $GW^0$  method for computing electronic excited states of molecules, SIAM Materials (Philadelphia, USA, May 2016)
- Error estimates for transport coefficients in molecular dynamics, SIAM Materials (Philadelphia, USA, May 2016)
- Error estimates on the computation of transport coefficients, workshop “Challenges in statistical mechanics: from mathematics to molecular dynamics to technological applications” (Imperial College London, UK, December 2015)
- Using Metropolis schemes to estimate correlation functions, meeting of the GdR ISIS (Telecom Paris, November 2015)
- Error estimates on the computation of transport coefficients, Program “Nonequilibrium Statistical Physics 2015” (ICTS, Bangalore, India, November 2015)
- Error estimates on the computation of transport coefficients, workshop NASPDE 2015 (Inria Sophia-Antiopolis, France, September 2015)
- Error estimates on the computation of transport coefficients, workshop “Free-energy calculations. A mathematical perspective” (BIRS at Oaxaca, Mexico, July 2015)
- Langevin dynamics with space-time periodic nonequilibrium forcing, workshop “Progress in nonequilibrium statistical mechanics” (Nice, France, June 2015)
- Energy (super)diffusion for systems with two conserved quantities, workshop “Analytic approaches to scaling limits for random systems” (HIM, Bonn, January 2015)

- Langevin dynamics with space-time periodic nonequilibrium forcing, 10th AIMS conference on Dynamical Systems, Differential Equations and Applications (Madrid, Spain, July 2014)
- An overview of numerical techniques for the simulation of quantum systems, workshop “Theoretical and Numerical Aspects of Quantum Transport” (Aalborg, Denmark, April 2014)
- Reponse of crystals to time-dependent perturbations, workshop “Mathematical and Numerical Analysis of Electronic Structure Models” (Berlin, April 2014)
- Error estimates in the numerical computation of transport coefficients, Oberwolfach meeting ”Large Scale Stochastic Dynamics” (Germany, October 2013)
- The microscopic origin of the macroscopic dielectric permittivity of crystals, QMaths12 (Berlin, September 2013)
- The microscopic origin of the macroscopic dielectric permittivity of crystals, GDRE ConEDP Meeting 2013 (Grenoble, April 2013)
- The microscopic origin of the macroscopic dielectric permittivity of crystals, Workshop on ”Quantum and Atomistic Modeling of Materials Defects” (IPAM, Los Angeles, October 2012)
- Time evolution of defects in crystals, Workshop on “Mathematical and Numerical Analysis of Electronic Structure Models” (Beijing, China, June 2012)
- Free energy techniques in Bayesian Statistics, CECAM workshop “Free energy calculations: From theory to applications” (Marne-la-Vallée, June 2012)
- Computation of transport properties by molecular dynamics, EPSRC Multiscale systems workshop (Warwick, UK, December 2011)
- Computation of transport properties by molecular dynamics, CECAM discussion on modeling of matter (Paris, December 2011)
- Nonequilibrium shear viscosity computations with Langevin dynamics, Minisymposium on mathematics in materials science, Beijing (China, September 2011)
- (Non)Equilibrium computation of free energy differences using Langevin dynamics, Molecular Kinetics, Berlin (September 2011)
- (Non)Equilibrium computation of free energy differences using Langevin dynamics, ICIAM, Vancouver (Canada, July 2011)
- Adiabatic switching for degenerate ground states, Summer school on Electronic Structure Analysis and Computation (Shanghai Jiao Tong University, June 2011)
- Thermal transport in one-dimensional systems: Some numerical results, Oberwolfach meeting ”Large Scale Stochastic Dynamics” (Germany, November 2010)
- (Non)Equilibrium computation of free energy differences using Langevin dynamics, Multiscale Molecular Modelling workshop (Edinburgh, United-Kingdom, June 2010)
- Computational statistical physics: a mathematical overview, Mathematical methods for ab-initio quantum chemistry (Nice, France, October 2009)
- Some nonlinear dynamics in computational statistical physics, ICNAAM 2009 (Rethymno, Greece, September 2009)
- Some adaptive dynamics in computational statistical physics, ICMS workshop “Adaptivity, robustness and complexity of multiscale algorithms” (Edinburgh, United-Kingdom, March 2009)
- Nonequilibrium computation of free energy differences: some new algorithms, Banff meeting on “Numerical methods for free energy computations” (Canada, June 2008)
- Computation of free energy differences, Oberwolfach meeting ”Atomistic models of materials” (Germany, April 2008)

- A reduced stochastic model for shock waves, IMA Summer Program Classical and quantum approaches in molecular modeling (Minneapolis, USA, August 2007)
- A simplified one-dimensional model for shock and detonation waves, Sixth Biennial International Conference on New models and hydrocodes for shock waves processes in condensed matter (Dijon, France, April 2006)
- (Non)equilibrium computation of equilibrium properties, ACI meeting - CIRM (Marseille, France, January 2006)
- Computing macroscopic properties using microscopic models, GdR CHANT meeting (Grenoble, France, January 2006)
- (Non)Equilibrium computation of free energy differences, Lake Arrowhead - IPAM culminating workshop (California, USA, December 2005)
- Workshop on Atomistic models and their continuum limits (Berlin, Germany, December 2003)

### Oral presentations in seminars

- Error estimates and variance reduction for nonequilibrium stochastic dynamics, LAMMe seminar, University of Evry (Evry, France, February 2026)
- Optimizing the diffusion for sampling with overdamped Langevin dynamics, Quantitative Methods for Finance seminar, University of Birmingham (Birmingham, United-Kingdom, October 2024)
- Importance sampling with free energies and autoencoders for multimodal probability distributions, Mostly Monte Carlo seminar (Paris, France, May 2024)
- Optimizing the diffusion for sampling with overdamped Langevin dynamics, Applied and Computational Mathematics Seminar of Heriot-Watt and University of Edinburgh (Edinburgh, United-Kingdom, February 2024)
- Enhanced sampling with autoencoders (Sanofi, Vitry-sur-Seine, December 2023)
- Quantifying the mini-batching error in Bayesian inference for Adaptive Langevin dynamics (Parisian seminar of statistics, March 2023)
- Coarse-graining and efficiently sampling with autoencoders, Materials Innovation Factory seminar (online, February 2023)
- Numerical methods for the linear response of nonequilibrium stochastic dynamics, Probability seminar, Université de Rennes (June 2022)
- Machine learning for coarse-graining molecular systems, Data Science and Computational Statistics Seminar, University of Birmingham (October 2021)
- Computation of transport coefficients in molecular dynamics: methods and numerical analysis, One World Stochastic Numerics and Inverse Problems seminar (online, July 2021)
- Hypocoercivity without changing the scalar product, Probability seminar, University of Bristol (online, May 2021)
- Longtime convergence of some diffusion processes in molecular dynamics, Probability seminar, University of Delaware (online, April 2021)
- Finding reaction coordinates with machine learning techniques for free energy computations, MPI Polymer Research public seminar (online, January 2021)
- Computation of transport coefficients in molecular dynamics: methods and numerical analysis, University of Massachusetts (April 2020)
- Linear response of nonequilibrium stochastic dynamics, ACM seminar, University of Edinburgh (February 2020)



- A mathematical introduction to molecular dynamics, MAC-MIGs students seminar Edinburgh (February 2020)
- Linear response of nonequilibrium stochastic dynamics, Probability group seminar, University of Bonn (January 2020)
- Convergence of Adaptive Langevin (and other dynamics) using hypocoercivity, Applied PDEs seminar Imperial College London (October 2019)
- Convergence of Adaptive Langevin (and other dynamics) using hypocoercivity, Université Paris-Dauphine (October 2019)
- Error estimates in molecular dynamics, DEFI/MEDISIM/POEMS seminar, Inria Saclay (July 2019)
- Hybrid Monte Carlo methods for sampling probability measures on submanifolds, Maths/chemistry seminar EMC2, Sorbonne-Université (June 2019)
- Convergence and approximation of Langevin like dynamics, MAP5 Colloquium, Université Paris Descartes (June 2019)
- Longtime convergence of evolution semigroups in molecular dynamics, Mathematical Physics Seminar, Institut Henri Poincaré (March 2019)
- Error estimates in molecular dynamics, Inria/LJLL seminar, Paris (December 2018)
- Hybrid Monte Carlo methods for sampling on submanifolds, Courant Institute of Mathematical Sciences, New York (October 2018)
- Sampling with stochastic differential equations: a primer, SMILE [Statistical Machine Learning in Paris] seminar, Paris (April 2018)
- Langevin dynamics at equilibrium and out of equilibrium: from hypocoercivity to efficient sampling, Duke University (February 2018)
- Mathematical analysis and numerical schemes for dissipative dynamics like models, ETH Zürich (February 2018)
- (Non)equilibrium Langevin dynamics: convergence and numerical approximation, University of Geneva (March 2017)
- (Non)equilibrium Langevin dynamics: convergence and numerical approximation, University of Massachusetts (February 2017)
- A mathematical introduction to some coarse-grained stochastic dynamics, Army Research Laboratory, Aberdeen Proving Ground (February 2017)
- Langevin dynamics with space-time periodic nonequilibrium forcing, University of Lille (September 2014)
- The computation of averages from equilibrium and nonequilibrium Langevin dynamics, AMMP seminar, Imperial College London (October 2013)
- Numerical simulation of Langevin dynamics, CEMRACS 2013 (August 2013)
- The microscopic origin of the macroscopic dielectric permittivity of crystals, Mathematical Physics Seminar, Institut Henri Poincaré (May 2013)
- Time evolution of defects in crystals, Collège de France (June 2012)
- Computation of transport properties by molecular dynamics, University of Edinburgh (February 2012)
- A mathematical understanding of the random phase approximation, Université de Cergy (October 2011)
- Adiabatic switching for degenerate ground states, Université de Lille (June 2009)

- Adaptive Importance Sampling (and applications to Bayesian statistics), BigMC seminar, Paris (June 2009)
- Adiabatic switching for degenerate ground states, IMA seminar on mathematics and chemistry, Minneapolis (May 2009)
- A mathematical introduction to statistical physics, Université de Strasbourg (February 2008)
- (Non)equilibrium computation of free energy differences, University of Warwick (February 2007)
- (Non)equilibrium computation of free energy differences, Seminar of the Computational Chemistry group at university of Amsterdam (June 2006)
- Out-of-equilibrium sampling, Meeting "Scientific computation" at université de Cergy (March 2006)
- A simplified one-dimensional model for shock and detonation waves, seminar Materials and Simulation Process Center (Caltech, November 2005)
- Theoretical and numerical comparison of some sampling methods in molecular dynamics, Presentation at the IPAM seminar (Los Angeles, September 2005)

### General audience talks and scientific outreach

- Chiche (presentation of research in numerical sciences for high students), 22 sessions (Mantes-la-Jolie, December 2023; Paris, March, May, November, December 2024; Palaiseau, March 2025; Paris, October 2025, November 2025, December 2025, January 2026)
- "AI improv sessions", Science feast Ecole des Ponts (Marne-la-Vallee, October 2025)
- "Flash science" session at the Double Science festival (Ground Control, Paris, June 2025)
- Presentation sessions for high school interns (Inria and Ecole des Ponts, June 2025)
- Presentation of research in mathematics for primary school students, 4 sessions (May 2024, Ecole Joliot-Curie, Champs-sur-Marne)
- "Predicting trajectories – from atom to galaxies", Science feast Ecole des Ponts (Marne-la-Vallee, October 2023)
- TV science show "On n'est pas que des cobayes" (France 5) on trajectories of fireworks (Paris, October 2015)
- TV science show "On n'est pas que des cobayes" (France 5) on pendulum waves (Paris, April 2015)
- Computer Simulations: The third way of doing science, ICMS (Edinburgh, June 2014)
- Les ordinateurs remplaceront-ils les expériences de laboratoire ?, Salon Culture et jeux mathématiques (Paris, Mai 2011)

### Posters

- Sampling constraints in average: The example of Hugoniot curves, Poster presented at CANUM, Carcans-Maubuisson, France (June 2010)
- Adaptive importance sampling strategies, poster presented at the IPAM workshop "Rare Events", UCLA (February 2009) and workshop "Molecular Simulations", IMA, Minneapolis (May 2009)
- Thermal transport in isotopically disordered carbon nanotubes: does the conductivity exist?, Poster presented at the Nanotubes'08 conference, Montpellier (July 2008)
- Local exchange potentials: A mathematical viewpoint, Poster presented at the Summer program "Classical and Quantum Approaches in Molecular Modeling", Minneapolis (2007)

- Path sampling with stochastic dynamics, Poster presented at the workshop “Sampling paths in molecular simulation: algorithms for phase transitions, reactivity and kinetics”, Orsay (november 2006)
- A simplified dual formulation of the electronic problem in terms of the second order reduced density matrix, Poster presented at the International Congress on Quantum Chemistry, Kyoto (may 2006)
- Equilibrium computation of free-energy differences using non-equilibrium methods, Poster presented at the workshops “Multiscale Modeling in Soft Matter and Bio-Physics” and “Time acceleration methods in molecular dynamics” at IPAM, Los Angeles (october 2005)

## Grants and contracts

|           |                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------|
| 2023-2026 | Sanofi contract on generative methods for drug design                                              |
| 2023-2026 | ICL/CNRS project (Nonequilibrium statistical physics: mathematical analysis and algorithms)        |
| 2023-2028 | project MAMABIO, PEPR B-BEST                                                                       |
| 2022-2025 | P.I. of ANR SINEQ (efficient simulation of nonequilibrium stochastic dynamics)                     |
| 2020-2023 | IFPEN contract on the development of machine learning techniques for catalytic mechanisms          |
| 2018-2021 | Sanofi contract on the development of machine learning techniques for finding reaction coordinates |
| 2016-2017 | P.I. of EARO contract on the development of numerical schemes for coarse-grained dynamics          |
| 2014-2017 | Contract with CEA/DAM on smoothed dissipative particle dynamics                                    |
| 2014-2018 | P.I. of ANR COSMOS (numerical techniques for computational statistics and molecular simulation)    |
| 2013-2016 | Contract with CEA/DAM on multiscale methods for the simulation of shock waves                      |

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|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| May 2026        | Co-organization of the workshop “Computational methods for probability distributions on manifolds” at IHP, Paris                                                                                |
| October 2025    | Co-organization of the closing workshop of ANR SINEQ at GSSI, L’Aquila                                                                                                                          |
| July 2025       | Co-organization of the CECAM research school “Sampling High-Dimensional Probability Measures with Applications in (Non)Equilibrium Molecular Dynamics and Statistics” at Birmingham, UK         |
| June 2025       | Co-organization of the workshop “Dimensionality reduction techniques for molecular dynamics” at ICMS, Edinburgh                                                                                 |
| March-May 2024  | Co-organization of the program “Data-Driven Materials Informatics” at IMSI, University of Chicago                                                                                               |
| September 2023  | Co-organization of the research school “Sampling high dimensional probability measures” (Ecole des Ponts)                                                                                       |
| June 2023       | Co-organization of the minisymposium “Numerical methods in statistical physics” at MCM2023 (Paris)                                                                                              |
| June 2022       | Co-organization of a CECAM workshop on “Chasing CVs using Machine Learning: from methods development to biophysical applications”                                                               |
| May 2021        | Co-organization of a mini-symposium at SIAM “Materials Science” 21 on “Computational Statistical Physics and Related Fields”                                                                    |
| 2018-2020       | Co-organization of the working group “Machine learning and optimization” (Labex Bezout)                                                                                                         |
| December 2018   | Co-organization of the CECAM discussion meeting “Coarse-graining with machine learning in molecular dynamics” (Gentilly, France)                                                                |
| November 2018   | Co-organization of the workshop “Computational Statistics and Molecular Simulation: A Practical Cross-Fertilization” (BIRS, Oaxaca)                                                             |
| September 2018  | Co-organization of the workshop “Advances in Computational Statistical Physics” (CIRM, France)                                                                                                  |
| April-July 2017 | Co-organization of the IHP Trimester “Stochastic dynamics out of equilibrium” (Paris, France)                                                                                                   |
| February 2016   | Co-organization of the workshop “Computational statistics and molecular simulation” (Paris, France)                                                                                             |
| June 2014       | Co-organization of the workshop “Computational methods for statistical mechanics - at the interface between mathematical statistics and molecular simulation” (ICMS, Edinburgh, United-Kingdom) |
| November 2012   | Co-organization of the workshop “Nonequilibrium Statistical Mechanics: Mathematical Understanding and Numerical Simulation” (Banff, Canada)                                                     |
| September 2012  | Co-organization of a CFCAM discussion on “Numerical methods and mathematical approaches for solar devices” (Paris, France)                                                                      |
| November 2011   | Co-organization of the workshop “Interactions between PDEs and probability theory” in the framework of the GdR CHANT (Grenoble, France)                                                         |
| May 2011        | Co-organization of the mini-symposium “Numerical methods in molecular simulation” (SMAI 2011, Guidel, France)                                                                                   |
| September 2010  | Co-organization of the workshop “Simulation of hybrid dynamical systems and applications to molecular dynamics” (IHP, Paris, France)                                                            |
| April 2008      | Co-organization of the workshop “Mathematical methods for molecular simulation” (HIM, Bonn, Germany)                                                                                            |
| 2005-2007       | Co-organization of the seminar “Scientific computing” at CERMICS, ENPC                                                                                                                          |

## Scientific responsibilities

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-              | Head of CERMICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2023-              | Member of the editorial board of <i>Journal of Computational Dynamics</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2022-              | Member of the Executive Board of GdR IAMAT (Artificial Intelligence and Materials Science)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 2021-2024          | Member of the Faculty Board of EELISA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2019-2024          | Elected member of the Teaching and Research Council of Ecole des Ponts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2015-2021          | Member of the scientific council of UNIT (Université Numérique Ingénierie et Technologie)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2014-2024          | Head of the team “Modeling, analysis and simulation” at CERMICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Journal refereeing | Adv. Comput. Math., AMRX, Ann. Appl. Probab., Ann. Henri Poincare, Ann. IHP Proba. Stat., Applied Math. Lett., Archive Rat. Mech. Anal., Bernoulli, Chem. Phys. Lett., Chem. Rev., Commun. Comput. Phys., Commun. Math. Phys., Commun. Math. Sci., Commun. Pure Appl. Math., Discrete Contin. Dyn. Syst. B, Entropy, Found. Comput. Math., ESAIM Proc., EPJ, J. Chem. Phys., ESAIM-Math. Model. Num., IMA J. Numer. Anal., J. Chem. Theor. Comput, Journal of Computational Dynamics, J. Comput. Phys., J. Mach. Learn. Res., J. Math. Phys., J. Maths Pure Appl., J. Roy. Stat. Soc. A & B, J. Stat. Phys., JSTAT, Lett. Math. Phys., Math. Comp., Mol. Simul., Nonlinearity, Phys. Chem. Chem. Phys., Phys. Rev. Lett., Phys. Rev. A, Phys. Rev. E, Physica D, PNAS, Probab. Theor. Rel. Fields, Proc. Roy. Soc. A, Pure and Applied Analysis, Questionaes Mathematicae, RapidRechLett, Sci. Rep., SIAM J. Appl. Math., SIAM J. Imaging Sciences, SIAM J. Appl. Dyn. Syst., SIAM Multiscale Model. Simul., SIAM J. Numer. Anal., SIAM Rev., SIAM J. Sci. Comput., SIAM JUQ, Stat. Comput., Stoch. Proc. Appl., Trans. Mach. Learn. Res. |
| Grant refereeing   | ERC (Europe), DFG (Germany), ISF (Israel), Millenium (Chile), NWO (Netherlands), RGC (Hong-Kong), ANR, Emergence Sorbonne-Université, CY Initiative (France)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## Evaluation panels

|             |                                                                                                     |
|-------------|-----------------------------------------------------------------------------------------------------|
| 2024        | Member of the HCERES evaluation jury of SAMM (Université Paris Panthéon)                            |
| Spring 2021 | Participation to the selection for PhD fellowships of the Udopia program at Université Paris-Saclay |



## Participation to recrutement juries

|             |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| Fall 2024   | Recruitment jury for research and teaching position at Ecole des Ponts           |
| Spring 2024 | Recruitment jury for professor position at Université d'Orléans                  |
| Spring 2023 | Recruitment jury for research position at Sorbonne Université                    |
| Spring 2023 | Recruitment jury for research position at Ecole des Ponts                        |
| Spring 2020 | Recruitment jury for associate professor position at Aalborg University, Denmark |
| Spring 2016 | Recruitment jury for research positions at Inria Paris                           |

## Participation to PhD juries

|                |                                                                                 |
|----------------|---------------------------------------------------------------------------------|
| June 2026      | PhD thesis of Francis Lörler, University of Bonn (referee)                      |
| February 2024  | PhD thesis of Iain Souttar, Heriot-Watt University (referee)                    |
| October 2023   | PhD thesis of Gong Chen, Sorbonne Université (referee)                          |
| October 2023   | PhD thesis of Théo Jaffrelot-Inizan, Sorbonne Université                        |
| September 2023 | PhD thesis of Thimotée Devergne, Sorbonne Université                            |
| November 2022  | PhD thesis of Paul Rohrbach, University of Cambridge (referee)                  |
| April 2022     | PhD thesis of Benjamin Stottrup, Aalborg University (referee)                   |
| March 2022     | PhD thesis of Lorenzo Campana, Inria Sophia (referee)                           |
| February 2022  | PhD thesis of Clovis Lapointe, Ecole Centrale Lille (referee)                   |
| December 2021  | PhD thesis of Lorenz Richter, BTU Cottbus (referee)                             |
| December 2021  | PhD thesis of Robert Benda, Ecole polytechnique                                 |
| July 2021      | PhD thesis of Louis Thiry, ENS Ulm (president)                                  |
| April 2021     | PhD thesis of Anne-Francoise Lempereur de Guerny, Sorbonne Université (referee) |
| February 2020  | PhD thesis of Anton Martinsson, University of Edinburgh (referee)               |
| December 2019  | PhD thesis of Nadia Jbili, Université Paris-Dauphine (referee)                  |
| Fall 2019      | PhD thesis of Nada Cvtekovic, FU Berlin (referee)                               |
| June 2019      | PhD thesis of Laurent Laffèche, Université Paris-Dauphine                       |
| June 2019      | PhD thesis of Nicolas Brosse, Ecole polytechnique (referee)                     |
| December 2018  | PhD thesis of Sabri Souguir, Université Paris Est                               |
| December 2017  | PhD thesis of Viviana Letizia, Université Paris Dauphine (referee)              |
| October 2017   | PhD thesis of Romain Poncet, Ecole polytechnique (referee)                      |
| June 2014      | PhD thesis of Marie Kopec, ENS Rennes (referee)                                 |

## Supervision of postdoctoral researchers

|           |                                           |
|-----------|-------------------------------------------|
| 2019-2020 | Geneviève Robin (funded by Inria Paris)   |
| 2020-2021 | Urbain Vaes (funded by FSMP)              |
| 2022-2023 | Mohamad Rachid (funded by ANR QuamProcs)  |
| 2024      | Hadrien Vroylandt (funded by PEPR B-BEST) |

## PhD supervisions

|           |                                                                                                                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2013-2016 | Ahmed-Amine Homman<br>Numerical integration of reduced stochastic dynamics for the simulation of shock waves                                                                                |
| 2013-2016 | Zofia Trstanova (co-supervised with Stéphane Redon, Inria Grenoble)<br>A mathematical analysis of some importance sampling strategies in molecular dynamics                                 |
| 2011-2017 | Alessandra Iacobucci (co-supervised with Stefano Olla; PhD work completed beside her full time job as research engineer)<br>Nonequilibrium stationary states of rotor and oscillator chains |
| 2014-2017 | Gérôme Faure<br>Multiscale methods for the simulation of shock and detonation waves                                                                                                         |
| 2015-2018 | Pierre Terrier<br>Numerical simulations for predicting the microstructural evolution of ferritic alloys. A study of Cluster Dynamics.                                                       |
| 2015-2018 | Julien Roussel<br>Theoretical and numerical analysis of non-reversible dynamics in computational statistical physics                                                                        |
| 2016-2019 | Grégoire Ferré (PhD prizes 2020 from Université Paris Est, and Ecole des Ponts)<br>Large deviations theory in statistical physics: some theoretical and numerical aspects                   |
| 2016-2019 | Lingling Cao (co-supervised with Eric Cancès)<br>Mathematical analysis of models of electronic structure for defected materials                                                             |
| 2018-2022 | Zineb Belkacemi (co-supervised with T. Lelièvre and E. Gkeka)<br>Deciphering protein function with artificial intelligence                                                                  |
| 2019-2022 | Inass Sekkat<br>Large scale Bayesian inference                                                                                                                                              |
| 2020-2023 | Thomas Pigeon (involved in the supervision by T. Lelièvre and P. Raybaud)<br>Rare event sampling methods and machine learning approaches for to study catalytic reaction mechanisms         |
| 2021-2024 | Renato Spacek (co-supervised with P. Monmarché; funding from FSMP)<br>Efficient computation of transport coefficients in molecular dynamics                                                 |
| 2021-2024 | Régis Santet (co-supervised with T. Lelièvre)<br>Foundations and optimization of Langevin samplers                                                                                          |
| 2022-2025 | Shiva Darshan (co-supervised with S. Olla)<br>Linear response of constrained stochastic dynamics                                                                                            |
| 2022-2025 | Noé Blassel (co-supervised with T. Lelièvre)<br>Mathematical and numerical approximation of quasi-stationary distributions                                                                  |
| 2023-     | Charlotte Chapellier (co-supervised with T. Lelièvre and E. Gkeka)<br>Generative methods for drug design                                                                                    |
| 2023-     | Pierre Marmey (involved in the supervision by T. Lelièvre and P. Raybaud)<br>Rare event sampling for catalytic reactions                                                                    |

|       |                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023- | Raphael Gastaldello (co-supervised with U. Vaes; ICL/CNRS fellowship)<br>Efficient numerical methods for Green–Kubo and Einstein like formulas |
| 2024- | Louis Carillo (involved in the supervision by T. Lelièvre and U. Vaes)<br>The narrow escape problem                                            |
| 2025- | Jean-Baptiste Himbert (co-supervised with T. Lelièvre)<br>Machine learning for molecular dynamics in materials science                         |
| 2025- | Antonin Coatantiec (co-supervised with T. Lelièvre)<br>Mathematical and numerical analysis for sampling metastable dynamics                    |

## Internship supervisions

|           |                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024      | Louis Carillo (ENS Saclay, 4 months), on microcanonical Langevin dynamics (supervised with U. Vaes)                                                                                |
| 2023      | Pierre Marmey (ENS Lyon, 5 months), on implementation of Adaptive Multilevel Splitting in OpenMM                                                                                   |
| 2022      | Noé Blassel (Sorbonne University, 6 months), on constrained stochastic dynamics in nonequilibrium systems                                                                          |
| 2021      | Remi Delloque (Ecole Normale Supérieure Lyon, 2.5 months), on hypocoercivity                                                                                                       |
| 2021      | Shiva Darshan (Ecole polytechnique, 4 months), on coupling techniques for sensitivity estimates                                                                                    |
| 2018      | Ghita El-Himdi (ENSTA, 2 months), on large scale Bayesian inference                                                                                                                |
| 2015      | Grégoire Ferré (ENPC, 6 months industrial internship), on machine learning techniques to compute potential energies in molecular dynamics                                          |
| 2013-2014 | Julien Roussel (ENPC, 1 year industrial internship), on the coupling of SPH and molecular dynamics                                                                                 |
| 2012-2013 | Gérôme Faure (ENPC, 1 year industrial internship), on density dependent potentials in molecular dynamics                                                                           |
| 2012-2013 | Pierre Terrier (ENPC, 1 year industrial internship), on numerical methods for the computation of shear viscosities                                                                 |
| 2012-2013 | Ahmed-Amine Homman (ENPC, 3rd year internship), on the computation of surface tension                                                                                              |
| 2012      | Yoann Robin (Paris 6, Master ANEDP, 4 months), on numerical analysis of periodic Schrödinger operators                                                                             |
|           | Theophile Lambert (ENPC, first year of training, 3 months), on numerical methods in nonequilibrium statistical physics                                                             |
| 2011      | Laura Da Silva (Paris 7, Master 1 in biology (AIV), 2 months)                                                                                                                      |
|           | Etienne Germain (ENPC, first year of training, 3 months), on numerical methods in computational statistical physics                                                                |
| 2009      | Vincent Dardel (UPMC, Master 1 in applied mathematics, 6 months)                                                                                                                   |
|           | Quentin Carbonneaux (Ecole des Ponts, first year of training, 3 months), on a numerical method to solve the Schrödinger equation                                                   |
| 2008      | Ronan Costaouec (ENPC, Master 2 in applied mathematics, 3 months), on canonical sampling at fixed entropy                                                                          |
|           | Sophia Chami, Pierre Lesouhaitier, Alexis De Maack and Dimitry Sculy (ENPC, first year of training, 3 months), on models and methods for heat transport in one dimensional systems |

|      |                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2007 | Emmanuel Maruani (ENPC, first year of training, 3 months), on a reduced model for shock waves                                                                                  |
|      | Nicole Spillane (ENPC, first year of training, 3 months), on effective dynamics                                                                                                |
| 2006 | Yann Tréguer (ENPC, first year of training, 3 months), on sampling techniques for molecular dynamics                                                                           |
|      | (with Eric Cancès) Moncef Elacheche (Orsay, Master in applied mathematics, 4 months), on the computation of Optimized effective potentials in quantum chemistry                |
| 2005 | (with Eric Cancès) Daniel Cai, Pierre-Emmanuel Beluche and Bertrand Fan (ENPC, first year of training, 3 months), electronic structure computations using Hartree-Fock methods |

## Teaching

|           |                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-     | “Introduction to Machine Learning”<br>Institut polytechnique Paris, M1 Applied mathematics and statistics                                                                        |
| 2023      | “Mathematics in action”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                               |
| 2018-2024 | “Statistics and Data Science”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                         |
| 2022-2023 | “Basics of analysis”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                                  |
| 2015-2022 | “Analysis and Scientific computing”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                   |
| 2015-     | “Introduction to computational statistical physics”<br>Master Mathématiques et application, spécialité Mathématiques et modélisation,<br>Sorbonne-Université (ex-ANEDP, Paris 6) |
| 2015-2018 | “Maths1 & Maths2” (lectures on numerical methods only)<br>Ecole des Mines, Paris                                                                                                 |
| 2016-2017 | “Fourier Analysis”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                                    |
| 2011-2015 | “Introduction to scientific computing”<br>Ecole des Mines, Paris                                                                                                                 |
| 2008-2015 | “Spectral theory and Fourier transform”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                               |
| 2006-2015 | “Scientific computing”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                                |
| 2006-2022 | “Statistical physics and quantum physics” (projects)<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                  |
| 2011-2014 | “Spectral theory of Schrödinger operators”, Master 2 at Université de Marne-la-Vallée                                                                                            |
| 2012      | “Analysis”<br>Ecole Nationale des Ponts et Chaussées, Marne-la-Vallée                                                                                                            |
| 2008-2010 | Molecular Simulation, Master SMCD                                                                                                                                                |
| 2005      | “Scientific computing and optimization”<br>ESIEE, Marne-la-Vallée                                                                                                                |

## Studies in pedagogy

- G. Buisson and G. Stoltz, La classe inversée à grande échelle en école d’ingénieur, Actes du colloque QPES 2017, 633-640
- J.-Y. Poitrat and G. Stoltz, Classe inversée en formation d’ingénieur, Actes du colloque QPES 2013, 795-801